Pattern Recognition and Machine Learning

Chapter 2: Probability Distributions

Reading Notes

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Bernoulli Distribution

Bernoulli Distribution:

$$\text{Bern}(x|\mu) = \mu^x(1-\mu)^{1-x}, \quad x \in \{0,1\}, \quad 0 \le \mu \le 1$$

Likelihood:

$$p(\mathcal{D}|\mu) = \prod_{n=1}^{N} \mu^{x_n}(1-\mu)^{1-x_n}, \quad \mathcal{D} = \{x_1, \ldots, x_N\}$$

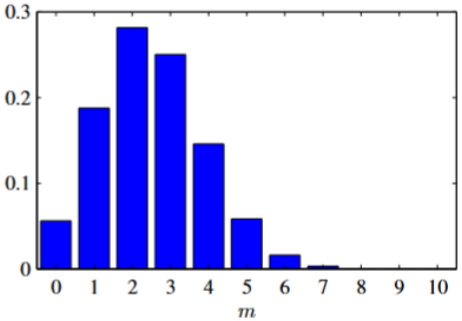
Log-likelihood:

$$\ln p(\mathcal{D}|\mu) = \sum_{n=1}^{N} \left[x_n \ln \mu + (1-x_n) \ln(1-\mu) \right]$$

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^{N} x_n$$

The log-likelihood depends on the data only through $\sum_{n=1}^{N} x_n$.

Binomial Distribution



Binomial distribution for $N = 10, \mu = 0.25$

Binomial Distribution:

$$\mathrm{Bin}(m|N,\mu) = \binom{N}{m}\mu^m(1-\mu)^{N-m}$$

where $\binom{N}{m} = \frac{N!}{m!(N-m)!}$.

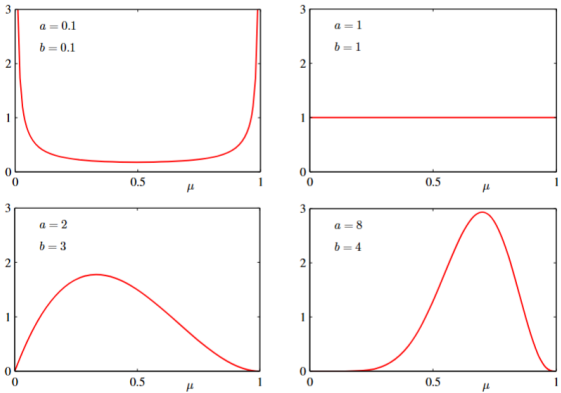
Properties:

$$\mathbb{E}[m] = N\mu$$
$$\mathrm{var}[m] = N\mu(1-\mu)$$

Note

The binomial is the distribution of the sum of N i.i.d. Bernoulli variables, so the sample mean $\frac{m}{N}$ is the maximum likelihood estimate of μ.

Beta Distribution (Conjugate Prior)



Beta distributions for various (a, b)

Beta Prior:

$$\text{Beta}(\mu|a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\mu^{a-1}(1-\mu)^{b-1}$$

- a, b: effective prior observations
- Mean: $\frac{a}{a+b}$, Variance: $\frac{ab}{(a+b)^2(a+b+1)}$

Posterior:

after observing m successes and l failures,

$$p(\mu|\mathcal{D}) = \text{Beta}(\mu|a+m, b+l)$$

Conjugacy

Beta is the conjugate prior for the binomial likelihood: Beta $\times$ Binomial $\rightarrow$ Beta

Predictive Distribution

Predictive distribution for the next trial:

$$p(x=1|\mathcal{D}) = \int_0^1 \mu\, p(\mu|\mathcal{D})\, d\mu = \mathbb{E}[\mu|\mathcal{D}]$$

Using the posterior $\text{Beta}(\mu|a+m, b+l)$:

$$p(x=1|\mathcal{D}) = \frac{a+m}{a+b+N}$$

- Interpolates between prior mean $\frac{a}{a+b}$ and sample mean $\frac{m}{N}$:

$$p(x=1|\mathcal{D}) = \lambda \cdot \frac{a}{a+b} + (1-\lambda) \cdot \frac{m}{N}, \quad \lambda = \frac{a+b}{a+b+N}$$

- As $N \to \infty$, tends to $\frac{m}{N} = \mu_{\text{ML}}$.
- Posterior variance decreases with N (on average).

Bayesian vs Frequentist

Bayesian results include prior knowledge and provide uncertainty estimates.

Multinomial Distribution

Categorical Distribution (1-of-K encoding):

$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{k=1}^{K} \mu_k^{x_k}, \quad \mathbf{x} = (0, \ldots, 1, \ldots, 0)^\top, \quad \sum_k \mu_k = 1, \quad \mu_k \geq 0$$

For N i.i.d. observations, the joint distribution over counts $\mathbf{m} = (m_1, \ldots, m_K)^\top$ is:

$$\text{Mult}(\mathbf{m}|\boldsymbol{\mu}, N) = \frac{N!}{m_1! \cdots m_K!} \prod_{k=1}^{K} \mu_k^{m_k}, \quad \sum_k m_k = N$$

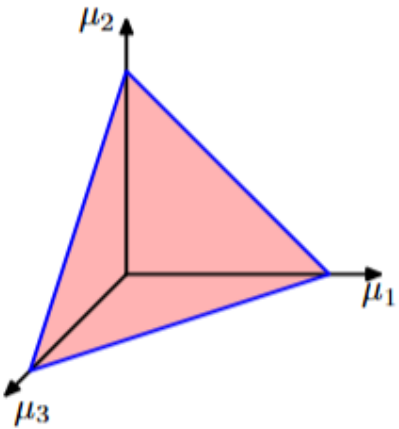
MLE:

$$\mu_k^{\text{ML}} = \frac{m_k}{N}, \quad \text{the fraction of observations in category } k$$

Sufficient Statistics

The likelihood depends on the data only through the counts $m_k = \sum_{n=1}^{N} x_{nk}$.

Dirichlet Distribution



The 2-simplex for $K = 3$: $\mu_1 + \mu_2 + \mu_3 = 1$

Dirichlet Prior:

$$\mathrm{Dir}(\boldsymbol{\mu}|\boldsymbol{\alpha}) = \frac{\Gamma(\alpha_0)}{\prod_{k=1}^{K}\Gamma(\alpha_k)}\prod_{k=1}^{K}\mu_k^{\alpha_k-1}, \quad \alpha_0 = \sum_{k=1}^{K}\alpha_k$$

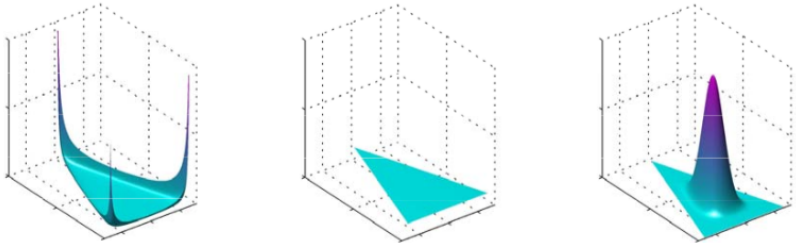
Support (Simplex): $\mu_k \geq 0$, $\sum_k \mu_k = 1$.

- α_k: effective prior counts for category k
- Mean: $\mathbb{E}[\mu_k] = \alpha_k/\alpha_0$

Posterior:
after observing counts $\mathbf{m} = (m_1, \ldots, m_K)$,

$$p(\boldsymbol{\mu}|\mathcal{D}) = \mathrm{Dir}(\boldsymbol{\mu}|\boldsymbol{\alpha} + \mathbf{m})$$

Dirichlet Density Shapes and Properties

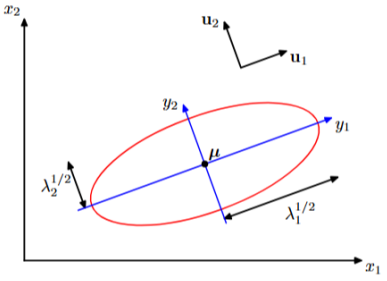


- $\alpha_k < 1$: density concentrated at corners (multi-modal, sparse)
- $\alpha_k = 1$: uniform distribution over the simplex (flat)
- $\alpha_k > 1$: density concentrated near the center (unimodal)

Key Properties

Dirichlet is conjugate to the multinomial: Dir $\times$ Mult $\rightarrow$ Dir.
For $K = 2$, Dirichlet reduces to the Beta distribution.

Multivariate Gaussian



Elliptical contours:

$(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = \text{const}$

Multivariate Gaussian:

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

- Mean: $\boldsymbol{\mu}$, Covariance: $\boldsymbol{\Sigma}$
- Mahalanobis distance: $\Delta^2 = (\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})$
- Linear transform:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b} \;\Rightarrow\; \mathbf{y} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b},\; \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$$

Geometric Interpretation

Eigen-decomposition: $\boldsymbol{\Sigma} = \sum_{i=1}^{D} \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$, $\boldsymbol{\Sigma}^{-1} = \sum_i \frac{1}{\lambda_i} \mathbf{u}_i \mathbf{u}_i^\top$

Conditional and Marginal Gaussians

Partition:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}$$

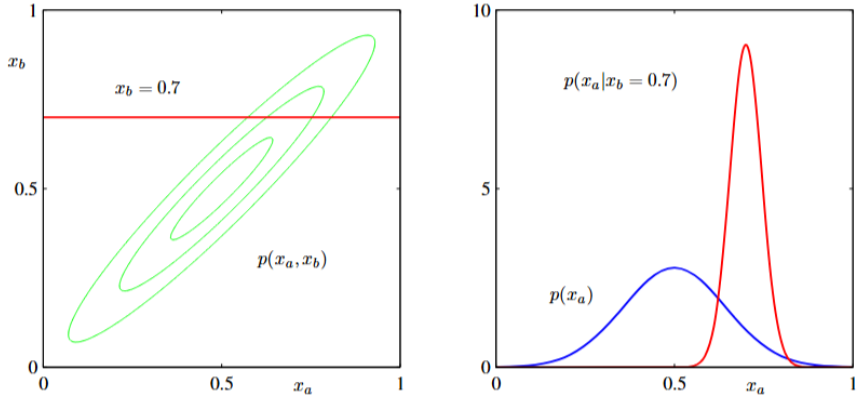
$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} \end{pmatrix}$$

	Mean	Covariance
Marginal $p(\mathbf{x}_a)$	$\boldsymbol{\mu}_a$	$\boldsymbol{\Sigma}_{aa}$
Conditional $p(\mathbf{x}_a \mid \mathbf{x}_b)$	$\boldsymbol{\mu}_a + \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}^{-1}(\mathbf{x}_b - \boldsymbol{\mu}_b)$	$\boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}^{-1}\boldsymbol{\Sigma}_{ba}$

Key Insights

- Both marginal and conditional distributions are Gaussian.
- Conditional mean is a **linear function** of $\mathbf{x}_b$; conditional covariance is **independent** of $\mathbf{x}_b$.

Joint, Marginal and Conditional Gaussians



Joint Gaussian $p(x_a, x_b)$ with marginal $p(x_a)$ (blue) and conditional $p(x_a|x_b)$ (red)

Joint Gaussian contours in (x_a, x_b) space

Marginal Integrate over x_b $\Rightarrow$ Gaussian $p(x_a)$

Conditional Fix x_b to a value $\Rightarrow$ Gaussian $p(x_a|x_b)$

Bayesian Inference for Gaussian (Known Variance)

Model: $p(\mathbf{x}|\mu) = \prod_n \mathcal{N}(x_n|\mu, \sigma^2)$.

Conjugate prior: $p(\mu) = \mathcal{N}(\mu|\mu_0, \sigma_0^2)$.

Posterior:

$$p(\mu|\mathbf{x}) = \mathcal{N}(\mu|\mu_N, \sigma_N^2)$$

where

$$\mu_N = \frac{\sigma^2}{N\sigma_0^2 + \sigma^2}\mu_0 + \frac{N\sigma_0^2}{N\sigma_0^2 + \sigma^2}\mu_{\mathrm{ML}}$$

$$\frac{1}{\sigma_N^2} = \frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}$$

Interpretation

Posterior precision = prior precision + data precision.

Bayesian Inference for Gaussian (Unknown Variance)

Conjugate prior for precision $\lambda = 1/\sigma^2$: Gamma distribution

$$\text{Gam}(\lambda|a,b) = \frac{b^a}{\Gamma(a)}\lambda^{a-1}e^{-b\lambda}$$

Posterior (after N observations):

$$\text{Gam}(\lambda|a_N,b_N)$$

with

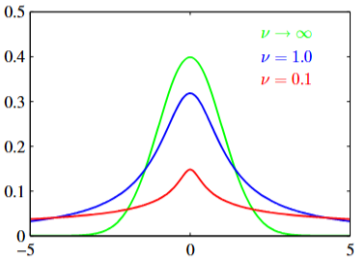
$$a_N = a_0 + \frac{N}{2}$$

$$b_N = b_0 + \frac{1}{2}\sum_{n=1}^{N}(x_n-\mu)^2$$

Interpretation

a and b correspond to effective prior observations and their sum of squares.

Student's t-Distribution



Student's t for various ν
($\nu \to \infty$: Gaussian)

$$\mathbb{E}[x] = \mu, \quad \nu > 1$$
$$\text{var}[x] = \frac{\nu}{\nu - 2}\lambda^{-1}, \quad \nu > 2$$

Definition (univariate):

$$\text{St}(x|\mu, \lambda, \nu) = \frac{\Gamma(\nu/2 + 1/2)}{\Gamma(\nu/2)} \left(\frac{\lambda}{\pi\nu}\right)^{1/2} \left[1 + \frac{\lambda(x-\mu)^2}{\nu}\right]^{-\nu/2-1/2}$$

Derivation: marginalizing a Gaussian over precision with Gamma prior.

$$p(x|\mu, \lambda, \nu) = \int_0^\infty \mathcal{N}(x|\mu, (\eta\lambda)^{-1}) \,\text{Gam}(\eta|\nu/2, \nu/2)\, d\eta$$

- Heavy tails $\Rightarrow$ robust to outliers.
- As $\nu \to \infty$, approaches Gaussian.
- Down-weights outliers naturally.

Mixtures of Gaussians

Motivation: Single Gaussian is unimodal and limited.

Mixture of K Gaussians:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \sum_k \pi_k = 1, \quad \pi_k \geq 0$$

Responsibilities:

$$\gamma_k(\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

Parameter Learning

Maximum likelihood has no closed form; use EM algorithm (Chapter 9).

Exponential Family Definition

General Form:

$$p(\mathbf{x}|\boldsymbol{\eta}) = h(\mathbf{x})g(\boldsymbol{\eta})\exp\{\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})\}$$

- $\boldsymbol{\eta}$: natural parameters.
- $\mathbf{u}(\mathbf{x})$: sufficient statistics.
- $g(\boldsymbol{\eta})$: normalization factor.

Examples:

- Bernoulli: $\eta = \ln \frac{\mu}{1-\mu}$, $u(x) = x$, $g(\eta) = \sigma(-\eta)$.
- Multinomial: $\eta_k = \ln \mu_k$, $u(\mathbf{x}) = \mathbf{x}$, $g = 1$ (with constraint) or softmax form.
- Gaussian: $\eta = (\mu/\sigma^2, -1/(2\sigma^2))^\top$, $u(x) = (x, x^2)^\top$.

Property

$\mathbb{E}[\mathbf{u}(\mathbf{x})] = -\nabla \ln g(\boldsymbol{\eta})$.

Maximum Likelihood and Sufficient Statistics

Likelihood for i.i.d. data:

$$p(\mathcal{D}|\boldsymbol{\eta}) = \left(\prod_n h(\mathbf{x}_n) \right) g(\boldsymbol{\eta})^N \exp \left\{ \boldsymbol{\eta}^\top \sum_n \mathbf{u}(\mathbf{x}_n) \right\}$$

MLE condition:

$$-\nabla \ln g(\boldsymbol{\eta}_{\mathrm{ML}}) = \frac{1}{N} \sum_n \mathbf{u}(\mathbf{x}_n)$$

The MLE depends on data only through $\sum_n \mathbf{u}(\mathbf{x}_n) \Rightarrow$ sufficient statistic.

Examples

Bernoulli: $\sum x_n$; Gaussian: $\sum x_n$ and $\sum x_n^2$.

Conjugate Priors in Exponential Family

Conjugate prior for likelihood:

$$p(\boldsymbol{\eta}|\boldsymbol{\chi}, \nu) = f(\boldsymbol{\chi}, \nu) g(\boldsymbol{\eta})^{\nu} \exp\{\nu \boldsymbol{\eta}^{\top} \boldsymbol{\chi}\}$$

Posterior:

$$p(\boldsymbol{\eta}|\mathcal{D}) \propto g(\boldsymbol{\eta})^{\nu+N} \exp\left\{\boldsymbol{\eta}^{\top}\left(\sum_{n} \mathbf{u}(\mathbf{x}_n) + \nu\boldsymbol{\chi}\right)\right\}$$

Same functional form $\Rightarrow$ conjugacy.

Interpretation: ν effective prior observations with sufficient statistic $\boldsymbol{\chi}$.

Examples

Beta for Bernoulli, Dirichlet for multinomial, Gaussian-Wishart for Gaussian.

Noninformative Priors

Goal: Let data speak for itself. Avoid subjective prior influence.

Location parameter $p(x|\mu) = f(x - \mu)$: Translation invariance $\Rightarrow$ $p(\mu) = \text{const}$ (improper).

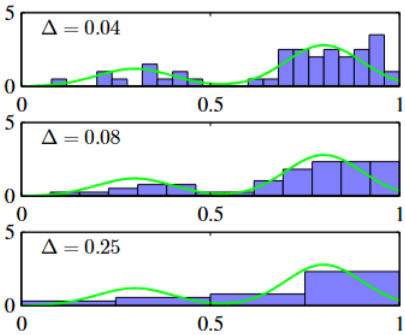
Scale parameter $p(x|\sigma) = \frac{1}{\sigma} f(x/\sigma)$: Scale invariance $\Rightarrow$ $p(\sigma) \propto 1/\sigma$ (improper), equivalently $p(\ln \sigma) = \text{const}$.

- Improper priors may lead to proper posteriors.
- Gaussian mean: uniform prior $\Rightarrow$ posterior $\mathcal{N}(\mu|\mu_{\text{ML}}, \sigma^2/N)$.
- Gaussian precision: $p(\lambda) \propto 1/\lambda$ (Jeffreys prior).

Caution

Noninformative priors are not invariant under reparameterization unless chosen properly.

Histogram Density Estimation



Histograms with different bin widths

Procedure: Partition space into bins of width Δ_i, count n_i points in bin i.

$$p_i = \frac{n_i}{N\Delta_i} \quad \Rightarrow \quad p(x) = \frac{n_i}{N\Delta_i} \text{ for } x \text{ in bin } i$$

- Simple, but discontinuous.
- Curse of dimensionality: number of bins grows exponentially.
- Bin width Δ controls smoothness (bias-variance trade-off).

Key Lessons

Density estimation relies on locality: we consider data points near the query point.
The bin width controls the bias-variance trade-off: too small gives noisy estimates, too large loses structure.

Kernel Density Estimation

General kernel estimator:

$$p(\mathbf{x}) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{h^D}k\left(\frac{\mathbf{x}-\mathbf{x}_n}{h}\right)$$

where $k(\mathbf{u}) \geq 0$, $\int k(\mathbf{u})d\mathbf{u} = 1$, h is bandwidth.

Common kernel: Gaussian

$$p(\mathbf{x}) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{(2\pi h^2)^{D/2}}\exp\left\{-\frac{\|\mathbf{x}-\mathbf{x}_n\|^2}{2h^2}\right\}$$

- Smooth and differentiable.
- Requires storing all data; $O(N)$ per evaluation.
- Bandwidth h is the main complexity parameter.

K-Nearest Neighbour Density Estimation

Procedure: Fix K. For query $\mathbf{x}$, find smallest sphere containing K points. Let its volume be V.

$$p(\mathbf{x}) = \frac{K}{NV}$$

- Adapts to local density (small V in dense regions).
- Not a proper density (integral diverges).
- K controls smoothness: $K = 1$ high variance / low bias; large K smoother / higher bias.

K-NN Classification

Assign test point to the majority class among its K nearest neighbours. $p(\mathcal{C}_k|\mathbf{x}) = K_k/K$, with K_k neighbours from class k.

Theory & Scalability

Cover-Hart: As $N \to \infty$, the 1-NN classifier has an error rate less than twice the Bayes error.
In practice: It requires $O(N)$ inference cost and storing all training data.

Summary

- **Bernoulli/Binomial** and **Beta** prior: for binary data.
- **Multinomial** and **Dirichlet** prior: for multi-class counts.
- **Gaussian**: multivariate distribution; conjugate priors for mean and precision.
- **Student's** t: heavy-tailed, robust to outliers.
- **Mixtures of Gaussians**: flexible multimodal densities.
- **Exponential family**: unifies many distributions; conjugate priors and sufficient statistics.
- **Nonparametric methods**: histograms, kernel density, K-NN; avoid parametric assumptions but suffer from the curse of dimensionality.

Key Takeaway

These distributions form the building blocks for probabilistic models and are essential for Bayesian inference and advanced techniques.